Secrets of Optical Flow- Solutions

1.1 Thought Experiments

Ans 1:

Optical flow can be used to create slow motion videos using frame interpolation. Frame interpolation means that given 2 frames, we would like to estimate a frame between those 2 frames.

We first estimate the optical flow between the frames of the given video to estimate the motion of pixels in each frame of the video. Using this optical flow, we can generate interpolated frames i.e. new frames that are in between the original frames. We can then merge the new interpolated frames between the original frames which will end up increasing the duration of each frame and make the motion appear slower.

Ans 2:

In the bullet time scene in the movie The Matrix, the action scene appears to be happening in slow motion. The technique used here is as follows

- The scene was captured by not one but multiple cameras positioned in a circular fashion.
- Each camera captured an image in quick succession.
- Using these sequences of images taken in quick succession, optical flow was estimated between consecutive frames.
- This optical flow was in turn used to generate interpolated frames between the original frames which were merged with the original frames thus giving a slow motion effect.

Ans 3:

Case-1: Sphere is rotating and light source is stationary

In this case motion field exists as the sphere is rotating about its axis. However, optical flow does not exist. This is because in a Lambertian Sphere the surface of the sphere is smooth with no texture and since the light source is fixed the reflective appearance of the sphere will remain the same.

Case-2: Sphere is fixed and light source is rotating

Here the motion field is zero but optical flow exists. This is because as the light source moves, the reflection appearance of the sphere would change and thus it would appear that pixels in the image are moving.

1.2 Concept Review

Ans 1:

The objective function in optical flow problem is based on the assumption that noise is absent. However, techniques such as median filtering help to deal with noise. Median filtering uses the median value of the pixels to replace the pixel values.

Ans 2:

In optical flow one of the 2 assumptions is that displacement (**δx,δy**) are small and the the time step **δt** is also small.

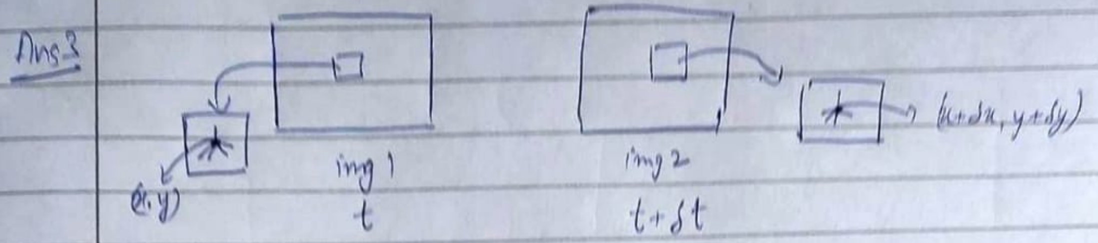
So in the Taylor series expansion:

$f(x + \delta x) = f(x) + (\partial f/\partial x)\delta x + (\partial^2 f/\partial x^2)(\delta x)^2/2! +$
$...(\partial^n f/\partial x^n)(\delta x)^n/n!$

The higher order terms can be neglected as (δx) is assumed to be small and therefore the higher order terms become negligible. This makes the equation of optical flow simpler to solve and the optical flow can be computed easily and faster. That is why this first-order Taylor series approximation is done.

Ans 3:

Ans 3



From the assumption 1 of optical flow we have

$$I(x, y, t) = I(x+\delta x, y+\delta y, t+\delta t) \quad \text{(brightness constancy)}$$

as $\delta x, \delta y, \delta t$ are small

$$I(x, y, t) = I(x, y, t) + \frac{\partial I}{\partial x}\delta x + \frac{\partial I}{\partial y}\delta y + \frac{\partial I}{\partial t}\delta t$$

(using Taylor Series approximation)

$$\Rightarrow \quad I_x \delta x + I_y \delta y + I_t \delta t = 0$$

$$\Rightarrow \quad I_x u + I_y v + I_t = 0$$

where $u = \frac{\delta x}{\delta t}$ is the x component of optical flow, $v = \frac{\delta y}{\delta t}$ is y component of optical flow

This is ~~an~~ the equation of a line in (u, v)

So ~~at~~ we have 2 variables (u, v) but only one equation. So (u, v) can't be computed as the optical flow estimation problem is an underconstrained problem. We know (u, v) lie on the line $I_x u + I_y v + I_t = 0$ but we don't know where.

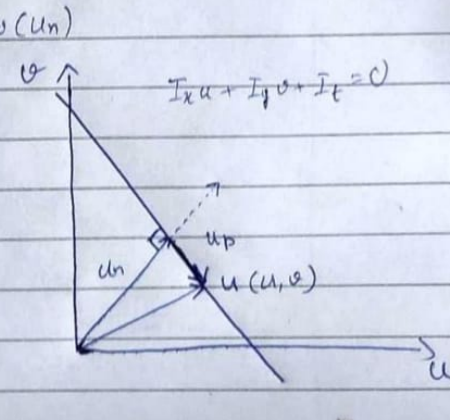
The optical flow u can split into normal flow (u_n) and parallel flow

$$u = u_n + u_p$$

Direction of normal flow: $\hat{u}_n = \frac{(I_x, I_y)}{\sqrt{I_x^2 + I_y^2}}$

Magnitude of normal flow $= \frac{|I_t|}{\sqrt{I_x^2 + I_y^2}}$

$$u_n = \frac{|I_t|}{\sqrt{I_x^2 + I_y^2}}(I_x, I_y)$$



However, u_p can't be determined (component ∥ to constraint line). As many combinations of (u, v) will satisfy the constraint equation, optical flow ~~is thus~~ constraint equation is ill posed.